ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
AIOT
04-05-2025

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Full Syllabus

ANSWER KEY — PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,B,C	A,C,D	A,B,C,D	A,C,D	A,B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	B	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	235.00 to 238.00	4.10 to 4.20	3.33	5.45 to 5.55	0.00	2.45 to 2.65	7.50	11.63 to 11.67

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C	C,D	A,C,D	A,B,C,D	A,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.80	2.50	140.00	4.00	13.85	12.00	75.00	20.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B	A,B	A,B,C	A,B,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	A	B	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.00	8.00	2.00	12.00	4.00	12.00	122.00	27.00

HINT – SHEET

For More Join: @JEEAdvanced_2025

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (C)

(P) **6.67°C**

Heat needed to melt ice = 20 × 80 = 1600 cal

Heat that water can provide by coming to

0°C = 40(1)(50) = 2000 cal

So, all ice melts and final temperature is between

0°C and 50°C

This temperature can be found by conservation of heat.

(Q) $50 \times 80 + 50 \times 1(T - 0) = 10 \times 540 + 10 \times (100 - T)$

$60T = 2400 \Rightarrow T = 40°C$

$100 \times 1 \times (100 - 50) = m\,(540)$

$m = \dfrac{5000}{540} = 9.25\text{gm}$

as $m < 10 \Rightarrow T_f = 100°C$

(S) $5 \times 540 + 5 \times 1 \times (100 - 0) + 16 \times 1(50 - 0)$

$= m(80)$

$4000 = 80m \Rightarrow m = 50$ gm

as $m < 60 \Rightarrow T_f = 0°C$

PART-1 : PHYSICS

SECTION-II

2. Ans (4.10 to 4.20)

According to the definition the heat capacity is written as

$$C = \frac{\delta Q}{\Delta T} = \frac{\Delta U + P\Delta V}{\Delta T} = C_V + P\frac{\Delta V}{\Delta T} \quad \text{.... (1)}$$

Let the gas parameters at the point A be equal (P_0, V_0, T_0). Then, the equation of the process takes the following form

$$\frac{V}{V_0} + \frac{T}{T_0} = 2 \qquad \text{... (2)}$$

For small deviations one derives

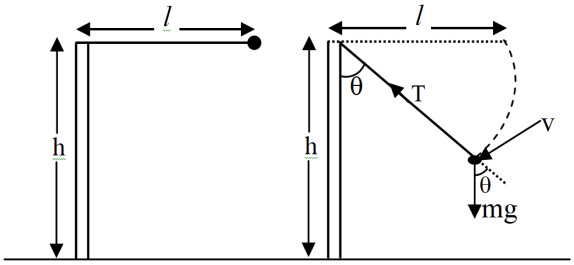
$$\frac{\Delta V}{V_0} + \frac{\Delta T}{T_0} = 0, \qquad \text{... (3)}$$

$$\frac{\Delta V}{\Delta T} = -\frac{V_0}{T_0} \qquad \text{... (4)}$$

With the aid of the equation of state $P_0V_0 = RT_0$, one finally gets

$$C = C_V + P\frac{\Delta V}{\Delta T} = C_V - P_0\frac{V_0}{T_0} = C_V - R = \frac{R}{2} \quad \text{...(5)}$$

7. Ans (7.50)



For bob conservation of mechanical energy :

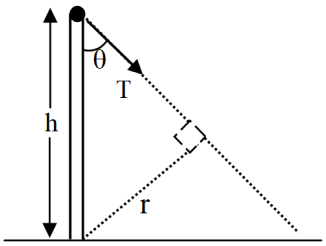
$$mg\ell\cos\theta = \frac{1}{2}mv^2$$

$$v = \sqrt{2g\ell\cos\theta} \quad \text{.....(1)}$$

Centripetal force : $T - mg\cos\theta = \dfrac{mv^2}{\ell}$

$$T = 3mg\cos\theta \quad \text{.....(2)}$$

Torque of tension about lower point of pole



$$\tau = Th\sin\theta$$

$$\tau = 3mgh\sin\theta\cos\theta$$

$$\tau = \frac{3}{2}mgh\sin 2\theta$$

$\therefore$ τ_{ext} should balance τ

$$\tau_{ext} = \frac{3mgh\sin 2\theta}{2}$$

$$\theta = 45^\circ \Rightarrow \tau_{ext_{max}} = \frac{3mgh}{2}$$

$$\tau_{ext_{max}} = \frac{3}{2} \times 0.5 \times 10 \times 1$$

$$\tau_{ext_{max}} = 7.50 \text{ Nm}$$

For More Join: @JEEAdvanced_2025

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (A,C,D)

at half of equivalence point.

pH = 11

$pOH = (14 - 11) = pK_b + \log 1$

$pK_b = 3$

5. Ans (A,D)

P =  Q = Br_2 (1 eq.) hν

R =  S = CH_3MgBr

T = 

U = (i) O_2, hν (ii) $H^{\oplus}$, Δ

6. Ans (B,C,D)

P = $CH_3-C\equiv CH$

Q = $CH_3-C\equiv\overset{\ominus}{C}Na^{\oplus}$

R = $CH_3-C\equiv C-CH_3$

S = 

T = $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH$

U = CHI_3

V = $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OCH_3$

W = $CH\equiv CH$

PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (C)

(P) $E_{cell} = 0.06\ (P^H anode - P^H cathode)$

pH Cathode = 3

for Anode $H_2CO_3 + Na_2CO_3 \longrightarrow 2NaHCO_3$

20 10 0

10 0 10

$pH = PK_{a1} = 7$

$E_{cell} = 0.06 \times (7 - 3) = 0.24$

(Q) for Anode $H_2CO_3 + NaOH \longrightarrow NaHCO_3$

20 20 20

0 0 40

pH = 9

$E_{cell} = 0.06 \times (9 - 3) = 0.36$

(R) for Anode $CH_3COOH + NaOH \longrightarrow CH_3COOH$

10 5 0

5 0 5

pH = 5

$E_{cell} = 0.06 \times (7 - 5) = 0.12$

(S) $P^H = 2$

$E_{cell} = 0.06 \times (7 - 2) = 0.3$

10. Ans (B)

Benzaldehyde do not give test with Fehling solution

* $NaHSO_3$ gives positive test with methyl ketone (not with Di ethyl ketone)

For More Join: @JEEAdvanced_2025

PART-2 : CHEMISTRY

SECTION-II

3. Ans (140.00)

Charge passed = 0.01 Faraday

At the anode $\left(H_2O \rightarrow \frac{1}{2}O_2 + 2H^+ + 2e^-\right)$ with 90 % efficientcy 0.01 × 0.9 F have been used and will produce $\frac{1}{4} \times 0.01 \times 0.9$ mole of O_2 i.e. 0.00225 mol O_2 .

At the cathode $2H_2O \xrightarrow{+2e^-} H_2 + 2OH^-$

moles of H_2 produced = $\frac{0.01 \times 0.8}{2}$ mol

= 0.004 mol

Total moles produced of gases

= 0.004 + 0.00225 = 0.00625 mol

vol. at STP = 0.00625 × 22400 mL = 140 mL

4. Ans (4.00)

$P(V - nb) = nRT$

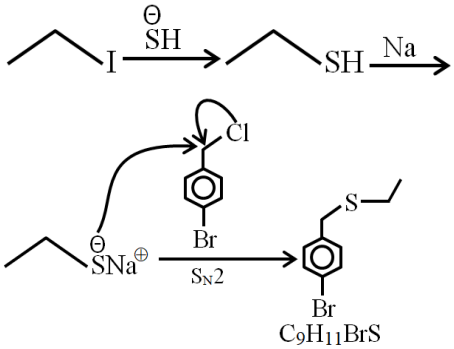
$\Rightarrow PV - nPb = nRT$

$\Rightarrow PV = nRT + nPb$ ..... (i)

$Z = \frac{PV}{nRT} = 1 + \frac{Pb}{RT}$ ..... (ii)

$\therefore \frac{b}{RT} = 0.01$ & $nRT = 40$

5. Ans (13.85)



% of sulphur in product

molecular weight of product = 231 ($C_9H_{11}BrS$)

% of S = $\frac{32}{231} \times 100 = 13.85$

6. Ans (12.00)

Total chiral centre = 4

Total stereo isomers = 8

$\therefore 4 + 8 = 12$

7. Ans (75.00)

Thymine contain 2 nitrogen atoms

Cytosine contain 3 nitrogen atoms

Guanine contain 5 nitrogen atoms

Adenine contain 5 nitrogen atoms

Nitrogen's in Thymine = 5 × 2 = 10

Nitrogen's in Cytosine = 5 × 3 = 15

Nitrogen's in Guanine = 5 × 5 = 25

Nitrogen's in Adenine = 5 × 5 = 25

Total nitrogen atoms = 75

8. Ans (20.00)

0.2 mole $K_4[Fe(CN)_6] \equiv$ 1.2 mole of C-atoms

33.33 % HCN is formed i.e. 0.4 mole of C-atoms are lost in the form of HCN.

So 0.8 mole CO is obtained.

$\underset{0.8\text{ mole}}{4CO} + \underset{\substack{0.2\text{ mole} \\ = 11.6\text{ gm}}}{Ni} \rightarrow \underset{\substack{0.2\text{ mole} \\ = 34\text{ gm}}}{Ni(CO)_4}$

100 gm Ni ⟶ 212 gm final weight

x gm ⟶ $\frac{212\,x}{100}$

So, $(x - 11.6) + 34 = \frac{212\,x}{100}$

x = 20.00 gm

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B,C,D)

$(2P - 3I)^4 = -216P$,

Take determinant on both sides we get

$|2P - 3I| = \pm 16 \times 27$

$\Rightarrow |2P - 3PP^{-1}| = \pm 16 \times 27$

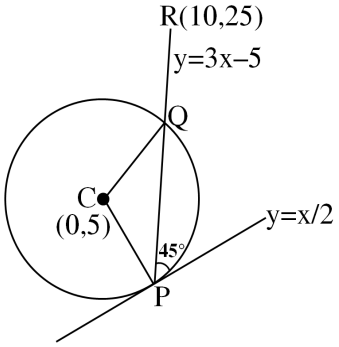
$|P|\,|2I - 3P^{-1}| = \pm 16 \times 27$

$|2I - 3P^{-1}| = \pm 27$

Now verify the options

For More Join: @JEEAdvanced_2025

2. Ans (A,B)



Here let slope of chord PQ = m ⇒ m = 3

Also it passer through R(10, 25)

∴ Equation of chord PQ $y = 3x - 5$

⇒ Radius of circle

$CP = \left|\frac{10}{\sqrt{5}}\right| = \sqrt{20}$

∴ Distance of chord PQ from C,

$d = \left|\frac{10}{\sqrt{10}}\right| = \sqrt{10}$

∴ length of chord $PQ = 2\sqrt{CP^2 - d^2} = 2\sqrt{10}$

Here Point P lies on line 2y = x and y = 3x – 5

∴ P ≡ (2, 1)

Now $Q \equiv \left(2 + 2\sqrt{10}\cos\theta, 1 + 2\sqrt{10}\sin\theta\right)$

Here $\tan\theta = 3 \Rightarrow \cos\theta = \frac{1}{\sqrt{10}}, \sin\theta = \frac{3}{\sqrt{10}}$

∴ Q ≡ (4, 7)

3. Ans (A,B)

$\sin6\beta + \sin2\beta = 3\cos4\beta$

$\Rightarrow 2\sin4\beta.\cos2\beta = 3\cos4\beta$

$\Rightarrow \frac{\sin4\beta.\cos2\beta}{\cos4\beta} = \frac{3}{2}$

$\Rightarrow \frac{\sin2\beta\cos^2 2\beta}{\cos4\beta} = \frac{3}{4}$

$\Rightarrow \frac{\sin(\alpha - 3\beta).\cos^2 2\beta}{\cos^2 2\beta - \sin^2 2\beta} = \frac{3}{4}$

$\Rightarrow \frac{2}{\sin\left(\frac{4\beta}{4}\right)} = \frac{1}{\cos\alpha} + \frac{1}{\sin\beta}$

$\Rightarrow \frac{1}{\sin\beta} = \frac{1}{\cos\alpha}$ ∴ $\sin\beta = \cos\alpha$

$\Rightarrow \cos\left(\frac{\pi}{2} - \beta\right) = \cos(5\beta)$

$\Rightarrow 5\beta = 2n\pi \pm \left(\frac{\pi}{2} - \beta\right)$ ∴ $\beta = (4n-1)\frac{\pi}{8}$

or $\beta = (4n+1)\frac{\pi}{12}$

4. Ans (A,B,C)

$15! = 2^{11}.3^6.5^3.7^2.11.13 = N$

Sum of all odd divisors

$(2^0)(3^0 + 3^1 + \ldots 3^6)(5^0 + 5^1 + 5^2 + 5^3)(7^0 + 7^1 + 7^2)$

$(11^0 + 11^1)(13^0 + 13^1)$

Product of all divisors $= N^{\frac{4032}{2}} = N^{2016}$

Product of all divisors which are not divisible by

$5 = \left(\frac{N}{5^3}\right)^{504}$

5. Ans (A,B,D)

$T_n = (a-3)n^3 + (b-4)n^2 + (a+b)n - 4$

for T_n of AP : a = 3, b = 4

$T_n = 7n - 4$

$d = T_n - T_{n-1}$

d = 7, first term = 3

$S_{10} = \frac{10}{2}(6 + 9 \times 7) = 345$

For More Join: @JEEAdvanced_2025

6. **Ans (A,B,C)**

$$\frac{a_n}{n} - \frac{a_{n+1}}{n+1} = n$$

$$\Rightarrow a_1 - \frac{a_{n+1}}{n+1} = \left(\frac{n(n+1)}{2}\right)$$

$$\Rightarrow 50 - n\left(\frac{n+1}{2}\right) = \frac{a_{n+1}}{n+1}$$

Now check

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (B)**

(P) $\tan(2\pi|\sin\theta|) = \tan\left(\frac{\pi}{2} - |\cos\theta|2\pi\right)$

$\Rightarrow |\sin\theta| + |\cos\theta| = \frac{n}{2} + \frac{1}{4}$ ...(1)

$\Rightarrow \sin\theta \quad 1 \le |\sin\theta| + |\cos\theta| \le \sqrt{2}$ then

$1 \le \frac{n}{2} + \frac{1}{4} \le \sqrt{2} \Rightarrow \frac{3}{2} \le n \le \frac{4\sqrt{2}-1}{2}$ then

$n = 2$ is only possible value.

$g(x) = \lim\limits_{n\to\infty}\left[2\left(\frac{4}{5}\right)^x\right] = 0$

(Q) $\tan^2\alpha + \cot^2\alpha$

$= \lim\limits_{x\to\infty}\left(\sqrt{x^2+8x+3} - \sqrt{x^2+4x+2}\right) = 2$

given $\tan^2\alpha + \frac{1}{\tan^2\alpha} = 2$

$\Rightarrow \tan^2\alpha = 1 \quad \Rightarrow \alpha = \pm\frac{\pi}{4}$

then $\frac{1}{2}(\tan\alpha + \cot\alpha) = 1, -1$

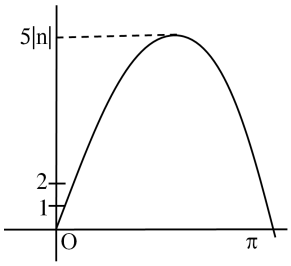
(R) $[2 + 5|n|\ |\sin x|] = 2 + [5|n|\sin x]$

Number of points of non derivability

$= 4(5|n| - 1) + 3 = 20|n| - 1 = 19$

$\Rightarrow |n| = 1$

(S) Adding equation (1) and (2)



$(\sin^{-1}y)^2 = \frac{4n+1}{32}\pi^2$

$\Rightarrow 0 \leqslant \frac{4n+1}{32}\pi^2 + \frac{\pi^2}{4} \Rightarrow \frac{-1}{4} \leqslant n \leqslant \frac{7}{4}$ ...(3)

Also $\cos^{-1}x = \frac{4n-1}{32}\pi^2 \Rightarrow \frac{1}{4} \leqslant n \leqslant \frac{8}{\pi} + 1$ ..(4)

From equations (3) and (4) $n = 1$

8. **Ans (A)**

$f(x) = \cos 2x$ and $g(x) = e^x$.

(I) $\lim\limits_{x\to0}(f(x))^{\frac{1}{x^2}} = e^{-2}$

(II) $\lim\limits_{x\to0}\frac{g(x) - f(x)}{2x} = \frac{1}{2}$

(III) $h(x) = e^x + 1$

$f(x)\ x \in (-\infty, 0]$, $h(x) \in (1, 2]$

(IV) $\cos 2x = \frac{1}{2}$ in $[0, 3]$ has 2 solutions

9. **Ans (B)**

(I) $OA = 1 + 4\cot\theta$

$OB = 4 + \tan\theta$

$OA + OB = 5 + 4\cos\theta + \tan\theta \ge 5 + 2\sqrt{4\cot\theta\tan\theta}$

(II) The reflection of P(4, −1) on y = x is Q(−1, 4)

Hence, $PQ = \sqrt{(4+1)^2 + (-1-4)^2} = \sqrt{50} = 5\sqrt{2}$.

(III) $AB = 2\sqrt{2}$

$OC = \sqrt{2}$

The maximum value of d is

$OF = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$

(IV) The given line is $x = y + 5$

Hence, the intercept made by the x-axis is 5.

For More Join: @JEEAdvanced_2025

10. Ans (C)

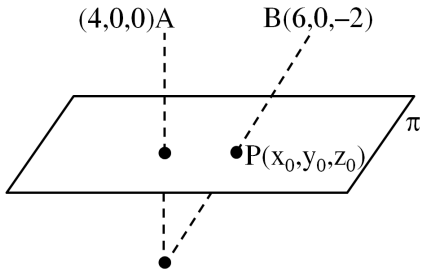
Let the plane be $\alpha x + \beta z + 1 = 0$

Pass through (1, 0, 1); (3, 2, −1)

$\therefore \alpha = \frac{1}{2}; \beta = -\frac{1}{2}$

(I) $\pi : x + z = 2$

(II)



Both A and B are on same side of π.

Reflection of A in plane is A'(2, 0, −2)

Equation of line $A'B = \vec{r} - = 6\hat{i} - 2\hat{k} + \lambda(4\hat{i})$

For P : $6 + 4\lambda + 0 - 2 = 2 \Rightarrow \lambda = \frac{-1}{2}$

$\therefore$ P(4, 2, −2)

$\therefore |4x_0 + y_0 + 2z_0| = 12$

(IV) Also, A' will lie on $\frac{x-2}{1} = \frac{y-\alpha}{0} = \frac{z+\beta}{-1}$.

$\Rightarrow \frac{2-2}{1} = \frac{0-\alpha}{0} = -\frac{2+\beta}{-1}$

$\Rightarrow \alpha = 0, \beta = 2$

$\therefore \alpha^4 + \beta^4 = 16$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.00)

$g(x) = \frac{f(x) - f(-x)}{|x| + 2} = \frac{8x^3 - 4x}{|x| + 2}$

$\therefore$ $g(x) = g(-x)$ $\therefore$ $g(x)$ is odd.

2. Ans (8.00)

At $x = 0, y = 4$

$e^x y = x^3 + 2x^2 + 4$

$\Rightarrow y''(0) = 8$

For $y''(0)$, $(e^{\sin x} - 1)^4 + (\cos x - 1)^3 e^{\sin x} + (e^x - e^{-x})^3$ is always gives zero at $x = 0$

3. Ans (2.00)

$0 \le \{x\} < 1$

$1 \le e^{\{x\}} < e$

4. Ans (12.00)

$\Delta = \frac{1}{2}.8.P = 4P$

$P = \left| \frac{3\cos\theta - 4 - \sin\theta}{\sqrt{2}} \right|$

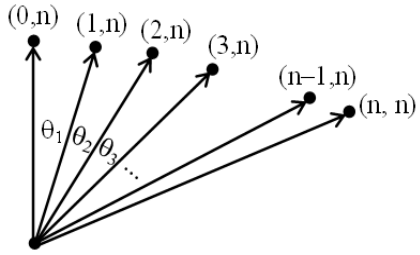
$2\sqrt{2}\left(4 - \sqrt{10}\right) = \Delta_1$

$2\sqrt{2}\left(4 + \sqrt{10}\right) = \Delta_2$

5. Ans (4.00)

For i = 0, 1, ..., n, define vectors $v_i = (i, n)$

Then, letting θ_{i+1} denote the angle between v_i and v_{i+1}, we have



$i(i+1) + n^2 = v_i \cdot v_{i+1}$

$= \sqrt{i^2 + n^2}\sqrt{(i+1)^2 + n^2}\cos\theta_{i+1}$

so $\theta_{i+1} = \text{arc cos}\left(\frac{i(i+1) + n^2}{\sqrt{i^2 + n^2}\sqrt{(i+1)^2 + n^2}}\right)$

Also, the sum $\theta_1 + \theta_2 + ... + \theta_n$ of all these angles is the angle between the y-axis and the line y = x,

so $\sum_{i=0}^{n-1}\theta_{i+1} = \frac{\pi}{4}$. Finally,

$\text{arc sin}\left(\frac{i(i+1) + n^2}{\sqrt{i^2 + n^2}\sqrt{(i+1)^2 + n^2}}\right)$

$= \frac{\pi}{2} - \text{arc cos}\left(\frac{i(i+1) + n^2}{\sqrt{i^2 + n^2}\sqrt{(i+1)^2 + n^2}}\right)$

Therefore,

$\sum_{i=0}^{n-1}\text{arc sin}\left(\frac{i(i+1) + n^2}{\sqrt{i^2 + n^2}\sqrt{(i+1)^2 + n^2}}\right) = \sum_{i=0}^{n-1}\left(\frac{\pi}{2} - \theta_{i+1}\right)$

$= \frac{n\pi}{2} - \frac{\pi}{4} = \frac{(2n-1)\pi}{4}$

For More Join: @JEEAdvanced_2025

6. Ans (12.00)

$a^2 = 13\lambda + r_1 \quad r_1 \in \{0, \pm1, \pm2, ..., \pm6\}$

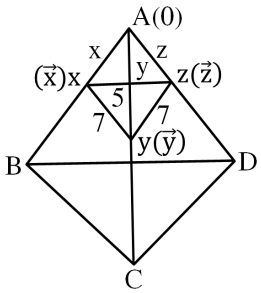
$b^2 = 13\lambda + r_2$

$c^2 = 13\lambda + r_3$

$$P = \frac{fav}{Total} = \frac{1 + 4 \times 3! + 4 \times 3! + 8 \times 3! + 4 \times 3! + 8 + 3!}{13^3}$$

$$P = \frac{169}{13^3} = \frac{1}{13}$$

7. Ans (122.00)



$|\vec{x}| = x$; $|\vec{y}| = y$; $|\vec{z}| = z$

Volume of tetrahedron AXYZ $= \frac{xyz}{6\sqrt{2}}$

$x^2 + y^2 - xy = 7^2$ ...(1)

$y^2 + z^2 - yz = 7^2$ ...(2)

$x^2 + z^2 - xz = 5^2$ ...(3)

On solving 3 equations we get

$xz = 12$, $x + z = \sqrt{61}$, $y = x + z$

Volume of tetrahedron $= \frac{1}{6\sqrt{2}} = \frac{xz(x+z)}{6\sqrt{2}} = \sqrt{122}$

8. Ans (27.00)

$43^{43} = (40 + 3)^{43} = 40k + 3^{43}$ ....(1)

$3^{43} = (3^2)^{21}.3 = (8 + 1)^{31}.3$

$= 4n + 3$ ....(2)

$\therefore 43^{43^{43}} = (40 + 3)^{4n+3} = 40k + 3^{4n+3}$

$= 3^{4n+3}$

$= (81)^n \times 27$

$= (80 + 1)^n \times 27$

$= 80k + 27$

$\therefore$ Remainder = 27

For More Join: @JEEAdvanced_2025